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Estimates of Direction and Intensity of the Palaeomagnetic Field from the Mugga Mugga Porphyry, Australia

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Summary

The mean direction of NRM in Mugga Mugga Porphyry (late Silurian or early Devonian intrusion near Canberra) is calculated to have declination 16° E and inclination −40° with Fisher precision estimate (k) equal to 43. This corresponds to a palaeomagnetic pole at 158° W, 71° N. Alternating field and thermal demagnetization, as well as divergence of the mean from the present field direction, indicate stability and that the NRM is primary thermoremanence. The intensity (M) of NRM which is destroyed by heating in zero field through a particular temperature range is compared with the intensity (M') of TRM acquired by cooling in the Earth's field through the same temperature range. By studying different temperature intervals, several estimates of M/M' were obtained from each specimen. If arbitrary correction is made for the effects of alteration above 550 °C, M/M' was found to be constant within a factor of about 2 over all temperature intervals in four samples. This suggests that serious decay of NRM has not occurred, and that M may be close to the original NRM intensity. Assuming intensity to be proportional to applied field, it is estimated that the Earth's field at the time when the Mugga Mugga Porphyry was intruded was approximately a quarter of its present intensity. A large number of estimates of M/M' is required before any estimate of ancient field intensity can be made; this is likely to apply to future studies of palaeointensity.

1. Introduction

During the preliminary palaeomagnetic work on Siluro-Devonian rocks in Australia (Irving & Green 1958, Green 1961) the direction of NRM was measured in nine samples from a quarry in Mugga Mugga Porphyry—a late Silurian (or possibly early Devonian) acid hypabyssal intrusion near Canberra (Öpik 1958) at 149°·4 E, 35°·1 S. In the present work, sampling was extended over much of the outcrop of the porphyry (Briden 1964) and the stability of NRM was tested by alternating field and thermal demagnetization. Because the NRM at the quarry is believed to be stable thermoremanence, attempts at determining ancient field intensity were made by comparing NRM with TRM imposed in the laboratory.

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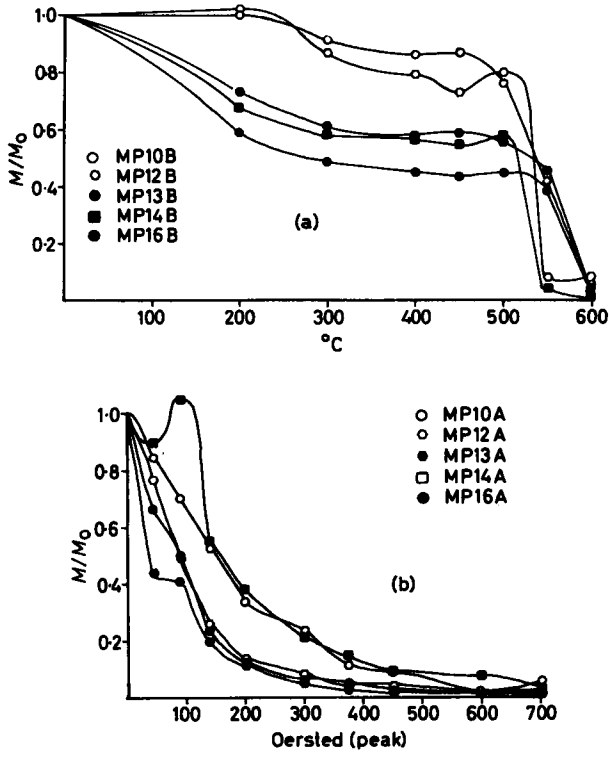


FIG. 1. Demagnetization of Mugga Mugga Porphyry from quarry, (a) by thermal treatment and (b) by alternating fields.

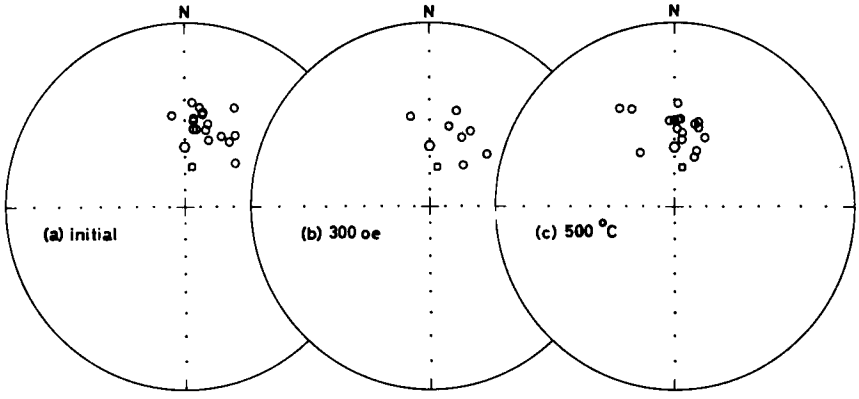


FIG. 2. Mean directions of NRM in samples of Mugga Mugga Porphyry from quarry. Present field direction is denoted by an open square and the geocentric axial dipole field by an open hexagon. Stereographic projection; all directions are inclined upward from the horizontal.

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Table 1

Directions of NRM in Mugga Mugga Porphyry from quarry before and after magnetic and thermal cleaning

Treatment	S	N	D	I	k	a	Pole ϕ	Pole λ
Initial	17	54	16	−40	43·3	5·5	158·2 W	71·4 N
300 oersted (peak)	7	7	22	−44	24·9	12·3	141·9 W	69·2 N
500 °C	17	45	2	−43	29·3	6·7	159·7 W	79·8 N
Initial (Green's samples)	a 9	11	26	−30	6·5	—	—	—
	b 9	11	29	−36	4·9	25·8	143·6 W	60·0 N

Number of samples (S), number of specimens (N), declination (D), inclination (I), measured positive downwards from horizontal, Fisher's precision parameter (k) and semi-angle of cone of confidence at 95% probability level (a) calculated by giving unit weight to each sample; palaeomagnetic pole position has longitude (ϕ) and latitude (λ).

a, Green's measurements; b, remeasured 3 years later by the author.

2. Direction and stability of NRM

Of a total of twenty-nine samples, seventeen were collected from the quarry faces (extending over 70 m horizontally and 25 m vertically) and twelve by blasting at natural outcrops. Directions of NRM were analysed by the method of Fisher (1953), giving unit weight to each sample (Table 1). Untreated NRM directions were well grouped about a mean which was only 25° away from the geocentric axial dipole field at Canberra, so that it was important to test the stability of NRM by laboratory methods. Maximum precision in a set of pilot specimens (Fig. 1) was obtained by partial alternating field and thermal demagnetization in 300 oersted (peak) and at 500 °C respectively; these optimum treatments were used for magnetic and thermal cleaning of additional specimens (Table 1). The estimates of precision obtained by analysing directions in all cleaned specimens are not significantly lower than the estimate from initial NRM directions (McElhinny 1964). Directions are illustrated in Fig. 2 and the corresponding palaeomagnetic pole positions, calculated on the assumption of a geocentric dipole field, are shown in Fig. 3. Initial intensities (M_0) of fifty four specimens from the quarry were in the range 0·12 to $1{\cdot}11 \times 10^{-3}$ Gauss with a mean value $0{\cdot}52 \times 10^{-3}$ Gauss. Initial susceptibility (K) of six specimens, measured in a field 0·54 oersted, ranged from 0·25 to $0{\cdot}97 \times 10^{-3}$ Gauss and the ratio M_0/K in these specimens ranged from 0·62 to 2·48 with an arithmetic mean value 1·32.

Directions at natural (blasted) outcrops were random with probability greater than 5% (Watson 1956). Both alternating field and thermal demagnetization failed to improve precision. Thermal demagnetization curves (Fig. 4) show a large thermally distributed component (Irving & Opdyke 1965) in contrast to the thermally discrete characteristic of the stable rocks (Fig. 6). Alternating field demagnetization curves of NRM and IRM imposed in 105 oersted in specimen MP31A were very similar. It is clear that NRM in natural outcrops is not identical to that in the quarries; in some cases it is likely to be of isothermal origin, and may be attributable to lightning, for several of the outcrops are on a prominent ridge. The NRM is very strong at one site which has been observed to have been struck by lightning. There is no indication that blasting has any effect on remanence in samples which are not shattered by the blast (Robertson 1963).

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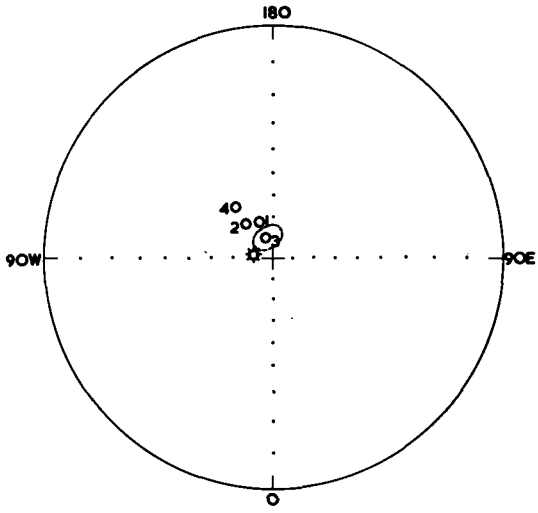


FIG. 3. Palaeomagnetic pole positions from Mugga Mugga Porphyry calculated from (1) initial directions of NRM, (2) after magnetic cleaning in 300 oersted peak field, (3) after thermal cleaning at 500 °C, showing the approximate oval of confidence at the 95% confidence level and (4) remeasurement of Green's samples. The geomagnetic pole is denoted by a star. Stereographic northern hemisphere projection.

The mean directions of NRM in the samples from the quarry which are given here are significantly different from the present and axial dipole field directions at Canberra (95% confidence); they are consistent with, but more precise than, Green's preliminary result. Thermal demagnetization of this NRM is similar to that of TRM in the laboratory (Fig. 6), and it is considered to be of thermoremanent origin. It is likely to be primary, because the intrusion is not thought to have been reheated and cooled through its Curie temperature subsequent to its emplacement and original cooling.

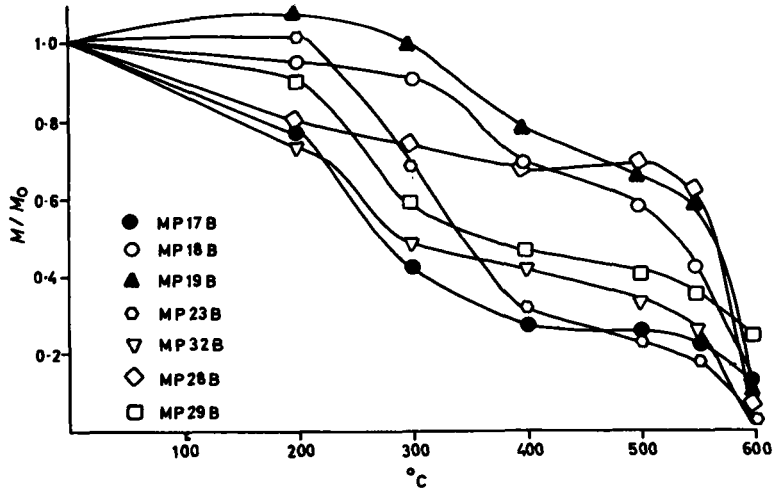


FIG. 4. Thermal demagnetization of Mugga Mugga Porphyry from natural outcrops.

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3. Basis of palaeointensity estimates

If the intensity of M' of TRM in rocks is related to the field F in which it was acquired, say

$$F = kM'^n$$

then, in principle, it is possible to calculate the intensity of the Earth's magnetic field in the geological past from measurements of NRM and TRM in unaltered igneous rocks in which the NRM may be assumed to be thermoremanence formed parallel to the Earth's field during cooling of the rocks. If NRM has not decayed with time, then ancient field intensity F_p should be given by

$$F_p = kM^n$$

where M is the intensity of NRM and k takes the same value as above. Hence

$$F_p = \left(\frac{M}{M'}\right)^n F$$

Néel's (1949) theory of single domain grains predicts that $n = \frac{1}{2}$. However, experimental evidence from a variety of rocks (Thellier 1938, Nagata 1943, Wilson 1961) suggests that intensity is proportional to field ($n = 1$) in fields of comparable intensity to the Earth's field, and this is compatible with later theories of thermoremanence (Néel 1955, Stacey 1958, Dickson 1962, Everitt 1961, 1962).

On the geocentric dipole model for the Earth's magnetic field, if $F = kM'$ and $F_p = kM$,

$$kM' = H_0(1+3\sin^2\lambda)^{\frac{1}{2}}$$

$$kM = H_{0p}(1+3\sin^2\lambda_p)^{\frac{1}{2}}$$

where H_{0p} is the palaeoequatorial intensity of the Earth's field and λ_p is the palaeomagnetic latitude of the place where the rock was formed. Hence

$$H_{0p} = \frac{M}{M'}\frac{F}{(1+3\sin^2\lambda_p)^{\frac{1}{2}}}$$

from which H_{0p} can be calculated.

Estimates of palaeointensity have been made by Thellier & Thellier (1959) using late Tertiary to recent lavas and archaeological specimens, and Wilson (1961), using baked Tertiary laterite, by comparing ancient and laboratory-imposed PTRM's. The procedure in the present experiments is essentially similar to that of previous workers, except that thermal analysis of NRM was carried out by thermal demagnetization in zero field and measurements were made at room temperature, whereas Thellier & Thellier had to carry out repeat experiments in a field after reversing the orientation of their specimens, and Wilson made measurements at high temperature during thermal demagnetization. The difference between intensity of NRM in rock specimens after demagnetization to T_1 °C and that after heating to a higher temperature T_2 °C is denoted by $M_{T_2-T_1}$; this is to be compared with $M'_{T_2-T_1}$ acquired by cooling the same specimens between T_2 and T_1 in the Earth's field, and in zero field outside that temperature range.

The prime obstacle to obtaining valid estimates of H_{0p} is alteration of rocks and their magnetic properties both in the field and in the laboratory. It is never possible to *prove* that a rock has remained petrologically unaltered since its emplacement, or that its NRM has not decayed with time. The best precaution is

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to use rocks which appear under the microscope to be unaltered, and of which the thermal demagnetization curves of NRM and TRM are similar over the range of the intensity measurements. The higher temperature PTRMs are, according to Néel's theories (1949, 1955), more stable with time than the lower temperature components, so that uniformity of NRM/TRM ratios in the same rock over several different temperature intervals, would suggest that decay had been unimportant; systematic variation would suggest that decay had occurred, and random variation would suggest that other factors were affecting the results or that measurements were not sufficiently precise. Another problem is to avoid contamination of the remanence by viscous and isothermal components, and spurious results due to stray fields in the furnace during cooling. Contamination was avoided by storing specimens in zero field after cooling through T_1 until measurement. The risk of oxidation of magnetite was minimized by heating and cooling in a nitrogen atmosphere. The effect of uniform stray fields was nullified by using sets of four specimens of the same sample in different orientations in the furnace, which also enables estimates of precision to be made. Checks against alteration were made at each stage by susceptibility measurements and by repeating lower temperature runs after the main experiments at higher temperature (changes in susceptibility or in low temperature PTRMs would indicate alteration).

Table 2

Sequence of detailed experiments on Mugga Mugga Porphyry

Run no.	Maximum temperature (°C)	Cooled in Earth's field in range (°C)	Cooled in zero field in range (°C)
1	Room temperature	(initial measurements)	
2	100	–	100–RT
3	100	100–RT	–
4	200	–	200–RT
5	200	200–100	100–RT
6	300	–	300–RT
7	300	300–200	200–RT
8	400	–	400–RT
9	300	300–100	100–RT
10	400	400–300	300–RT
11	450	–	450–RT
12	450	450–400	400–RT
13	500	–	500–RT
14	300	300–100	100–RT
15	500	500–400	400–RT
16	550	–	550–RT
17	550	550–500	500–RT
18	600	–	600–RT
19	300	300–100	100–RT
20	600	600–500	500–RT

(RT = room temperature)

4. Description of intensity experiments and results

Four non-oriented samples of Mugga Mugga Porphyry were marked so that relative orientation of specimens from the same sample was known. Four cylinders and two or three disks were used from each sample. A series of experiments (Table 2) was carried out to determine $M_{T_2-T_1}$ (runs 1, 2, 4, 6, 8, 11, 13, 16, 18) and $M'_{T_2-T_1}$ (runs 3, 5, 7, 10, 12, 15, 17, 20) over various temperature intervals. Runs 9, 14 and 19 were checks against alteration, by evaluating $M'_{300-100}$ after heating to 400 °C,

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500 °C, and 600 °C respectively, and comparing results with $M'_{300-100}$ calculated from runs 4–7. Intensity and direction in each specimen were measured after each run. Susceptibility in disks was measured initially and after all demagnetization runs, but these are only approximate because the a.c. bridge had to be used at very low sensitivity, which could not be reset accurately on successive occasions.

The partial thermal components of NRM and TRM over each temperature interval were calculated by a vectorial method, using the additive law of TRM; for example, for MP44A1

Run	Components measured	Intensity $\times 10^{-6}$ Gauss	Direction D	I	Principal components X	Y	Z
6	M_{T_c-300}	431·4	314	+59	155·4	−162·9	367·9
8	M_{T_c-400}	436·8	315	+57	168·5	−171·1	364·8
10	$M_{T_c-400}+M'_{400-300}$	383·3	322	+47	205·4	−157·9	282·4
by subtracting 6 from 8							
	$M_{400-300}$	15·7	148	+11	−13·1	8·2	3·1
by subtracting 8 from 10							
	$M'_{400-300}$	91·2	20	−65	36·9	13·2	−82·4

from which $M/M' = 0{\cdot}172$ (see Table 3). There is a wide spread of values of M/M' but there is no systematic difference between cylinders and disks, either in this ratio or in M and M' separately. Since cylinders and disks had different surface area, superficial alteration is discounted, and an analysis of mean intensity, direction and dispersion within a sample is performed on results from four cylinders from each sample. Mean direction is calculated by Fisher analysis of unit vectors; this avoids undue weighting of any specimen with anomalous intensity and gives an estimate of directional precision. Mean intensity is calculated vectorially. When precision is high, the mean direction of unit vectors will be indistinguishable from the mean of vectors which have length proportional to intensity. Precision is important because it is thought that, in an isotropic rock which took a long time to cool, each component $M_{T_2-T_1}$ should lie close to the direction of M_{T_c-RT} allowing a certain dispersion attributable to secular variation; each component $M'_{T_2-T_1}$ should lie close to the field direction in the laboratory. Accordingly, any $M_{T_2-T_1}$ which is grossly different in direction from that of the total NRM is not used in calculation of M/M' (Table 4). The criterion which has been adopted for rejection of such results is randomness with 1% probability (Watson 1956). All $M'_{T_2-T_1}$ were identical with the Earth's direction with 99% probability.

The measurements of $M'_{300-100}$ after heating to 300 °C, 400 °C, 500 °C and 600 °C (Fig. 5) showed that an important change in the capacity to acquire remanence occurred between 500 °C and 600 °C, and this conclusion is not contradicted by the observation that susceptibility falls after heating to 550 °C. In calculating H_{0p}, an adjustment is made to those estimates of M/M' obtained from temperature ranges which extend into the region where alteration is known to take place (taken as 550 °C). The supposition is made that if alteration causes $M'_{300-100}$ to increase by a factor C, then the proper value of $M'_{T_2-T_1}$ (T_1 greater than the temperature of alteration) is given by dividing the observed intensity by C (Table 5). $M'_{T_2-T_1}$, in this context, is used as a measure of the magnetic susceptibility of magnetically hard grains. This susceptibility is directly related to the remanence. This adjustment is only applied in eleven of the twenty-nine estimates, and brings them into broad consistency with the remainder.

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Table 3

M/M′ calculated for all temperature ranges

Spec. No.	400–200	300–100	600–400	500–300	200–100	100–RT	400–300	300–200	550–450	500–400	450–400	600–500	550–500	500–450	600–550
Cylinders															
MP44A1	0·029	0·234	0·276	0·091	0·736	0·448	0·296	0·172	0·519	0·259	0·085	0·161	1·220	0·064	0·087
MP44A2	0·266	0·319	0·051	0·102	0·527	1·019	0·257	0·669	0·343	0·174	0·099	0·601	0·571	0·117	0·101
MP44B1	0·221	0·209	0·091	0·069	1·248	0·524	0·490	0·074	0·140	0·122	0·075	0·145	1·081	0·237	0·067
MP44B2	0·652	0·113	0·063	0·104	0·458	1·365	0·225	0·106	0·195	0·164	0·102	0·466	0·469	0·076	0·104
MP45A1	0·519	0·175	0·140	0·120	0·400	1·161	0·224	0·221	0·189	0·122	0·121	0·207	0·313	0·116	0·124
MP45A2	0·488	0·435	0·214	0·067	0·606	0·483	0·401	0·368	0·165	0·086	0·085	0·108	0·368	0·079	0·091
MP45B1	0·561	0·375	0·194	0·128	0·682	0·738	0·452	0·339	0·154	0·121	0·129	0·260	0·338	0·104	0·137
MP45B2	1·120	0·332	0·223	0·099	0·296	3·617	0·130	0·527	0·176	0·115	0·108	0·270	0·715	0·098	0·099
MP46A1	0·313	0·382	0·097	0·073	0·605	0·421	0·268	0·561	0·202	0·061	0·082	0·145	0·407	0·101	0·083
MP46B1	0·350	0·245	0·309	0·122	0·174	0·709	0·296	0·242	0·431	0·176	0·118	0·454	1·449	0·130	0·117
MP46B2	0·203	0·176	0·221	0·090	0·555	0·291	0·190	0·339	0·183	0·095	0·085	0·117	0·278	0·089	0·048
MP46C1	0·406	0·123	0·134	0·094	0·488	0·841	0·405	0·206	0·125	0·105	0·095	1·169	0·377	0·084	0·102
MP47A1	0·403	0·085	0·119	0·132	0·525	0·634	0·348	0·457	0·149	0·134	0·131	0·188	0·229	0·126	0·196
MP47A2	0·176	0·113	0·111	0·075	0·620	0·406	0·621	0·551	0·354	0·114	0·062	0·377	0·343	0·094	0·003
MP47B1	0·241	0·081	0·137	0·119	0·565	1·342	0·349	0·391	0·168	0·126	0·114	0·181	0·175	0·123	0·070
MP47B2	0·436	0·214	0·136	0·096	0·693	0·625	1·298	0·456	0·184	0·117	0·090	0·424	0·104	0·103	0·017
Disks															
MP44A3	0·389	0·199	0·181	0·083	1·208	0·669	0·189	0·404	0·381	0·072	0·076	0·110	1·967	0·196	0·092
MP44A4	0·194	0·166	0·096	0·083	0·724	1·152	0·184	0·176	0·062	0·100	0·084	0·178	0·102	0·103	0·084
MP45A3	0·370	0·176	0·089	0·118	0·613	0·364	0·405	0·292	0·109	0·136	0·119	0·161	0·688	0·132	0·119
MP45A4	0·811	0·631	0·375	0·136	1·004	0·839	0·765	0·540	0·272	0·173	0·131	0·289	0·320	0·123	0·123
MP45B4	0·580	0·308	0·191	0·095	1·277	0·441	0·826	0·248	0·343	0·152	0·089	0·211	0·519	0·076	0·090
MP46A2	0·542	0·170	0·093	0·080	0·600	1·379	0·769	0·125	0·207	0·085	0·076	0·215	0·218	0·081	0·049
MP46A3	0·177	0·145	0·053	0·096	0·650	0·637	0·133	0·237	0·074	0·134	0·101	0·278	0·179	0·129	0·010
MP46C2	0·259	0·123	0·106	0·076	0·449	0·702	0·176	0·212	0·082	0·075	0·076	0·264	0·199	0·085	0·087
MP47A3	0·639	0·231	0·097	0·087	0·961	5·091	0·218	0·295	0·112	0·115	0·098	0·185	0·169	0·109	0·029
MP47B3	0·155	0·162	0·157	0·107	0·652	0·774	0·387	0·147	0·178	0·120	0·098	0·012	0·278	0·107	0·047

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Table 4

Sample mean intensities M, M′ and ratio M/M′ for all 'significant' determinations of M (see text). Intensities in Gauss ($\times 10^{-6}$)

Temperature range (°C)	MP44			MP45			MP46			MP47		
	M	*M′*	*M/M′*	*M*	*M′*	*M/M′*	*M*	*M′*	*M/M′*	*M*	*M′*	*M/M′*
300–100	—	—	—	85·5	137·0	0·624	—	—	—	—	—	—
400–200	—	—	—	47·8	153·4	0·312	33·7	161·2	0·209	—	—	—
500–300	—	—	—	39·7	222·0	0·179	—	—	—	19·9	159·2	0·125
600–400	431·0	4572·5	0·091	433·0	4062·5	0·107	277·4	3002·5	0·092	150·4	1474·5	0·102
100–RT	—	—	—	—	—	—	—	—	—	—	—	—
200–100	—	—	—	61·0	55·7	1·095	—	—	—	—	—	—
300–200	—	—	—	—	—	—	—	—	—	—	—	—
400–300	—	—	—	21·3	69·7	0·306	—	—	—	—	—	—
500–400	—	—	—	24.0	152·2	0·158	—	—	—	20·6	121·8	0·169
550–450	50·9	291·8	0·174	133·3	1220·0	0·109	114·4	1225·0	0·093	137·4	1132·0	0·121
600–500	414·7	4597·5	0·090	430·0	3865·0	0·111	268·0	2852·5	0·094	127·3	1328·5	0·096
450–400	—	—	—	—	—	—	—	—	—	—	—	—
500–450	—	—	—	20·3	53·0	0·383	—	—	—	15·8	81·5	0·194
550–500	—	—	—	114·5	1169·7	0·098	105·5	707·0	0·149	122·2	1114·2	0·110
600–550	381·5	4372·5	0·087	304·0	3190·0	0·095	177·7	1677·3	0·106	—	—	—

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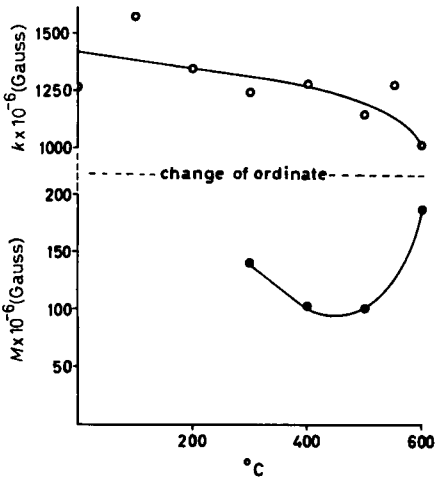


FIG. 5. Variation of susceptibility (K) and intensity $M'_{300-100}$ with maximum temperature of experiments, for sample MP45. The increase in intensity and fall in susceptibility between 550 °C and 600 °C are common to all samples, but the variations at lower temperatures are not.

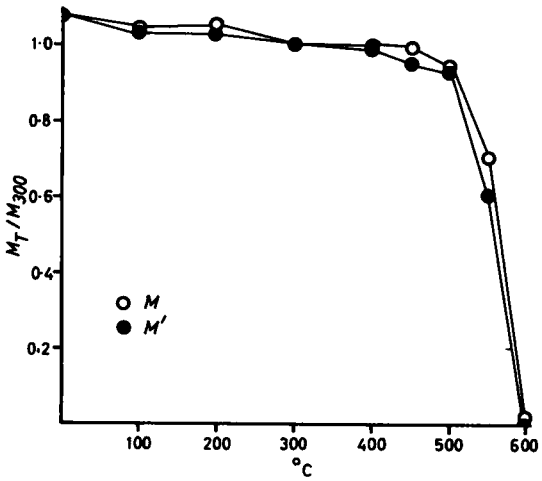


FIG. 6. Composite thermal demagnetization curves for NRM and TRM in all samples which were used, normalized to 300 °C to isolate low temperature effects.

Table 5

A check against alteration. $M'_{300-100}$ determined after experiments have proceeded to temperatures higher than 300 °C. *Intensities* $\times 10^{-6}$ *Gauss*

	Maximum temperature				
Sample	300 °C	400 °C	500 °C	600 °C	C
MP44	143·3	113·8	114·0	229·7	1·80
MP45	137·0	101·5	98·5	216·9	1·93
MP46	153·3	108·5	108·6	172·7	1·40
MP47	54·5	49·4	61·7	152·5	2·76

C = ($M'_{300-100}$ after 600 °C) divided by (average of three previous values of $M'_{300-100}$ taken from table)

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Table 6

*Estimates of H_{0p} using factor C where applicable (denoted by *), see text*

Temperature range (°C)	H_{0p} oersted			
	MP44	MP45	MP46	MP47
300–100	—	0·0635	—	—
400–200	—	0·1307	0·0877	—
500–300	—	0·0749	—	0·0525
600–400*	0·0686	0·0862	0·0540	0·1178
100–RT	—	—	—	—
200–100	—	0·4588	—	—
300–200	—	—	—	—
400–300	—	0·1229	—	—
500–400	—	0·0779	—	0·0708
550–450	0·0633	0·0881	0·0547	0·1395
600–500*	0·0679	0·0888	0·0547	0·1086
450–400	—	—	—	—
500–450	—	0·1604	—	0·0815
550–500	—	0·0793	0·0874	0·1270
600–550*	0·0653	0·0767	0·0624	—

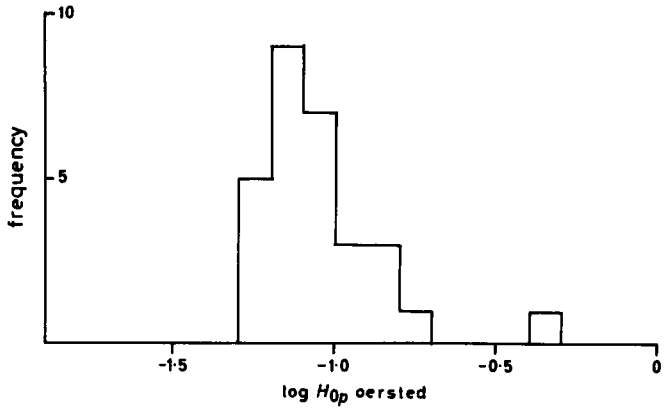


FIG. 7. Frequency histogram of estimates of H_{0p} from Mugga Mugga Porphyry, on logarithmic (base 10) scale.

5. Discussion

The estimates of equatorial field intensity (H_{0p}) (Table 6) may be analysed in terms of a lognormal distribution (Fig. 7) which gives a mean value of 0·087 oersted with standard error corresponding to the range 0·080 to 0·094, compared with its present value of 0·312 oersted. It is evident from this spread of estimates that a large amount of data is required in order to make a single justifiable estimate of ancient field intensity. Although the possibility that alteration and decay of the original NRM have taken place cannot be ruled out, the consistency of the estimates from different temperature intervals suggests that decay has not been important, and this is supported by the similarity of the thermal decay curves of NRM and TRM (Fig. 6). If the assumption that intensity is proportional to field is correct, the estimate that

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the intensity of the Earth's field at the time of intrusion of the Mugga Mugga Porphyry was about 28% of its present value may be reasonably correct. If estimates from temperature ranges below 550 °C and are used (i.e. not using the adjustment C) the mean is calculated to be 0·096 oersteds which is about 31% of the present field strength.

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